THE UNIVERSITY OF AZAD JAMMU AND KASHMIR, MUZAFFARABAD



COURSE TITLE	Software Construction & Development
COURSE CODE	SE-3101
LAB TITLE	Use Case Diagram
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SUBMISSION DATE	October 26, 2026
LAB NUMBER	01

1. Objective

The objective of this lab is to understand and practise requirements modelling with UML use case diagrams. After completing this lab, the student is able to:

- Identify the actors that interact with a software system.
- Identify the use cases (functional requirements) that a system must provide.
- Show relationships between actors and use cases (association, include and extend).
- Draw a complete use case diagram with a system boundary for a real-world scenario.

2. Background

A use case diagram is a UML behavioural diagram that describes what a system does from the point of view of its users, without describing how it does it internally. The notation used in this report is summarised below.

Element	Meaning
Actor (stick figure)	A person or external system that interacts with the system. A primary actor starts a use case; a secondary actor (for example a payment gateway or a card machine) supports it.
Use case (oval)	A single goal or function that the system performs for an actor.
System boundary (rectangle)	Separates what is inside the system from the actors outside it.
Association (solid line)	Connects an actor to a use case it takes part in.
<<include>> (dashed arrow)	The base use case always uses the behaviour of the included use case. The arrow points to the included use case.
<<extend>> (dashed arrow)	An optional use case adds behaviour to a base use case under some condition. The arrow points to the base use case.

Tools used: draw.io (diagrams.net) / StarUML for drawing, and Microsoft Word for this report.

3. Question 1: Online Ticket Reservation System

Task: Using knowledge of how a ticket is reserved online, develop a use case diagram that could serve as a basis for understanding the requirements of an online ticket reservation system.

3.1 Assumptions

The task does not name a type of ticket, so the system is kept general (bus, train, flight or event). The following assumptions are made:

- A customer must have an account and be logged in before booking.
- Payment is done online through an external payment gateway.
- After a successful booking, the system sends a confirmation by email or SMS and the customer can download the e-ticket.
- An administrator maintains schedules, fares and users, and views reports.

3.2 Actors

Actor	Type	Role
Customer	Primary	Searches for tickets, reserves and pays for them, cancels bookings and downloads e-tickets.
Administrator	Primary	Manages the schedules, fares and user accounts, and generates reports.
Payment Gateway	Secondary (external)	Authorises and processes the online payment and refund requests from the system.

3.3 Use Cases

#	Use Case	Description
1	Register Account	A new customer creates an account by providing name, email, phone and password.
2	Login	A registered user is authenticated before using the booking features.
3	Search Available Tickets	The customer searches by origin, destination, date and class and sees available options.
4	Select Seat	The customer chooses seat(s) from the seat map. It is part of every booking.
5	Book / Reserve Ticket	The main use case. The customer confirms the selected trip and passenger details to reserve the ticket.
6	Make Payment	The fare is paid online through the payment gateway. It is part of every booking.
7	Send Confirmation (Email/SMS)	The system sends the booking confirmation and reference number.

#	Use Case	Description
8	Cancel Reservation	The customer cancels an existing booking.
9	Process Refund	If the cancelled ticket is refundable, the refund is processed back to the customer.
10	View Booking History	The customer views past and upcoming bookings.
11	Download / Print E-Ticket	The customer downloads or prints the ticket for travel.
12	Manage Schedules	The administrator adds, edits or removes routes, trips and timings.
13	Manage Fares	The administrator sets ticket prices and discounts.
14	Manage Users	The administrator views, blocks or removes user accounts.
15	Generate Reports	The administrator produces sales and booking reports.

3.4 Relationships

- **Book / Reserve Ticket <<include>> Select Seat:** a booking cannot be completed without choosing a seat.
- **Book / Reserve Ticket <<include>> Make Payment:** every booking must be paid for.
- **Book / Reserve Ticket <<include>> Send Confirmation:** a confirmation is always sent after booking.
- **Process Refund <<extend>> Cancel Reservation:** a refund happens only if the cancelled ticket is refundable, so it is optional behaviour.

3.5 Use Case Diagram

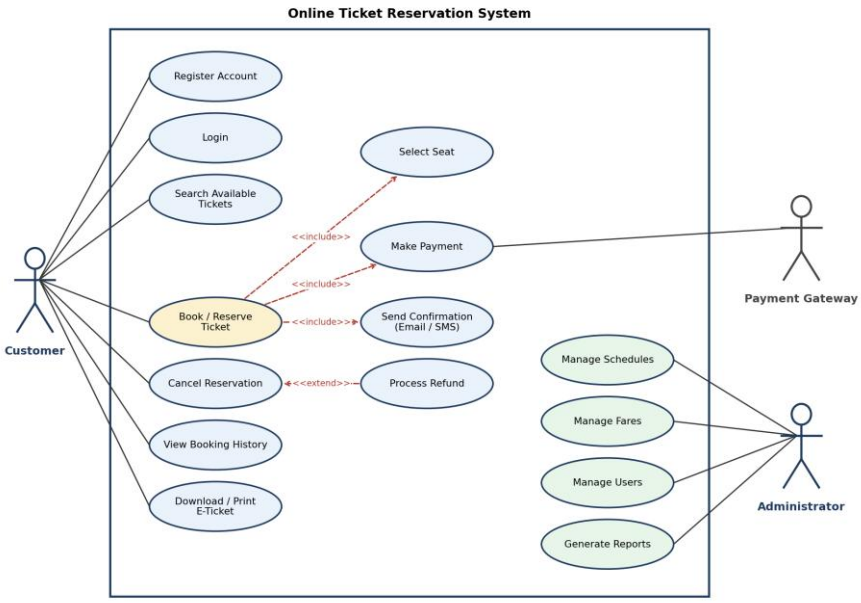


Figure 1: Use case diagram of the Online Ticket Reservation System

3.6 Explanation

The rectangle represents the boundary of the Online Ticket Reservation System. Everything inside is done by the system; the actors stay outside.

The Customer is linked to the use cases he or she starts directly: Register Account, Login, Search Available Tickets, Book / Reserve Ticket, Cancel Reservation, View Booking History and Download / Print E-Ticket. Select Seat is not linked to the customer directly because it is always performed as part of booking, so it is shown with an <<include>> relationship.

Book / Reserve Ticket is the central use case. It includes Select Seat, Make Payment and Send Confirmation, which means those three steps are always carried out whenever a ticket is booked. Make Payment is linked to the Payment Gateway because the gateway is an external system that actually performs the transaction.

Process Refund extends Cancel Reservation. A normal cancellation may simply end, but when the ticket is refundable the refund behaviour is added, which is why <<extend>> is used instead of <<include>>.

The Administrator, shown on the right, is linked to the management use cases: Manage Schedules, Manage Fares, Manage Users and Generate Reports. These keep the data that customers search and book against up to date.

4. Question 2: Credit Card Processing System

Task: Identify the actors and use cases of the given case study and draw the use case diagram.

4.1 Case Study Summary

A customer buys an item from a shop in a mall using a credit card. The vendor prepares the bill and swipes the card. The credit card machine (card reader) processes the transaction, checks the card limit and updates the balance, and prints a balance statement/receipt. The vendor gives the statement to the customer, who signs the receipt and returns it to the vendor.

4.2 Actors

Actor	Type	Role
Customer	Primary	Selects the item, hands over the credit card, requests the transaction amount and signs the receipt.
Vendor	Primary	Prepares the bill, swipes the card, issues the balance statement and collects the signed receipt.
Credit Card Machine	Secondary (system / card reader)	Processes the transaction, checks and updates the card balance and prints the receipt.

Note: the bank that owns the card account is the party behind the card machine. It is not named in the case study, so it is not drawn as a separate actor. If the instructor wants it, it can be added as a secondary actor connected to Verify Credit Limit and Update Card Balance.

4.3 Use Cases

#	Use Case	Description
1	Select Item	The customer chooses the item to buy from the shop.
2	Prepare Bill	The vendor makes the bill for the selected item.
3	Hand Over Credit Card	The customer gives the credit card to the vendor for swiping.
4	Request Transaction Amount	The customer states the amount to be charged to the card.
5	Swipe Card	The vendor swipes the card in the credit card machine.
6	Process Transaction	The card reader carries out the required amount transaction.
7	Verify Credit Limit	The system checks that the amount does not exceed the card's limit/balance.
8	Update Card Balance	The card balance is updated after a successful transaction.
9	Print Balance Statement / Receipt	The credit card machine prints the balance statement or receipt.
10	Issue Balance Statement	The vendor gives the printed statement to the customer.
11	Sign Receipt	The customer signs the receipt.
12	Collect Signed Receipt	The vendor takes the signed receipt back from the customer.

4.4 Relationships

- **Swipe Card <<include>> Process Transaction:** swiping the card always triggers the transaction.
- **Process Transaction <<include>> Verify Credit Limit:** the limit is always checked before approval.
- **Process Transaction <<include>> Update Card Balance:** a processed transaction always updates the balance.
- **Issue Balance Statement <<include>> Print Balance Statement / Receipt:** the vendor can only issue a statement after the machine has printed it.

4.5 Use Case Diagram

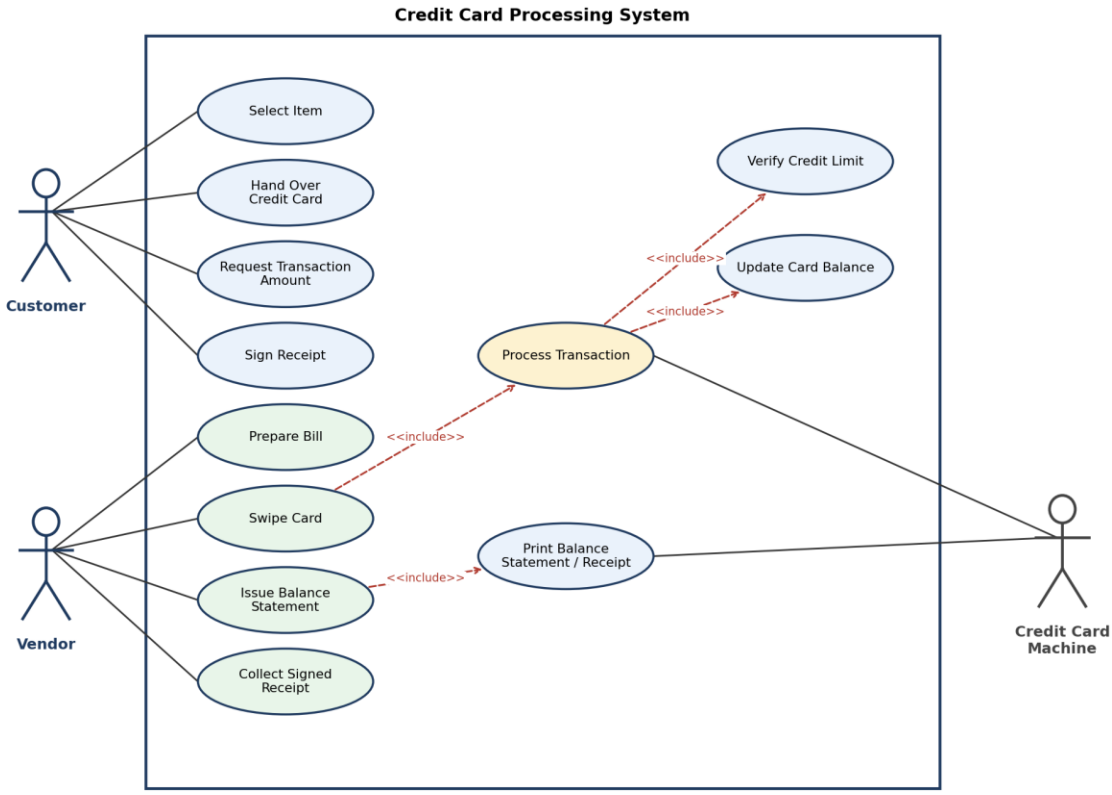


Figure 2: Use case diagram of the Credit Card Processing System

4.6 Explanation

The system boundary represents the Credit Card Processing System. The Customer and the Vendor are placed on the left because they are the human actors, and the Credit Card Machine is on the right as a secondary (system) actor.

The Customer is associated with Select Item, Hand Over Credit Card, Request Transaction Amount and Sign Receipt. The Vendor is associated with Prepare Bill, Swipe Card, Issue Balance Statement and Collect Signed Receipt. The Credit Card Machine is associated with Process Transaction and Print Balance Statement / Receipt, because these are done by the card reader itself.

The main flow follows the case study in order: the customer selects an item; the vendor prepares the bill; the customer hands over the card and requests the amount; the vendor swipes the card; the machine processes the transaction (verifying the credit limit and updating the balance); the machine prints the statement; the vendor issues it; the customer signs it; and the vendor collects the signed receipt.

Swipe Card includes Process Transaction, and Process Transaction includes Verify Credit Limit and Update Card Balance. These are <<include>> relationships because each of them always happens and none is optional. Issue Balance Statement includes Print Balance Statement / Receipt, because the statement must be printed by the machine first.

6. Conclusion

In this lab, two use case diagrams were developed. For the Online Ticket Reservation System, three actors and fifteen use cases were identified, with Book / Reserve Ticket as the central use case that includes seat selection, payment and confirmation, and Process Refund as an extension of cancellation. For the Credit Card Processing System, three actors and twelve use cases were identified, covering the full flow from item selection to the signed receipt, with the transaction processing broken into credit limit verification and balance update.

The lab showed how a use case diagram turns a plain description of a system into a clear set of actors, goals and relationships that can be used as the starting point for requirements and design.